ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Cloud MS Azure

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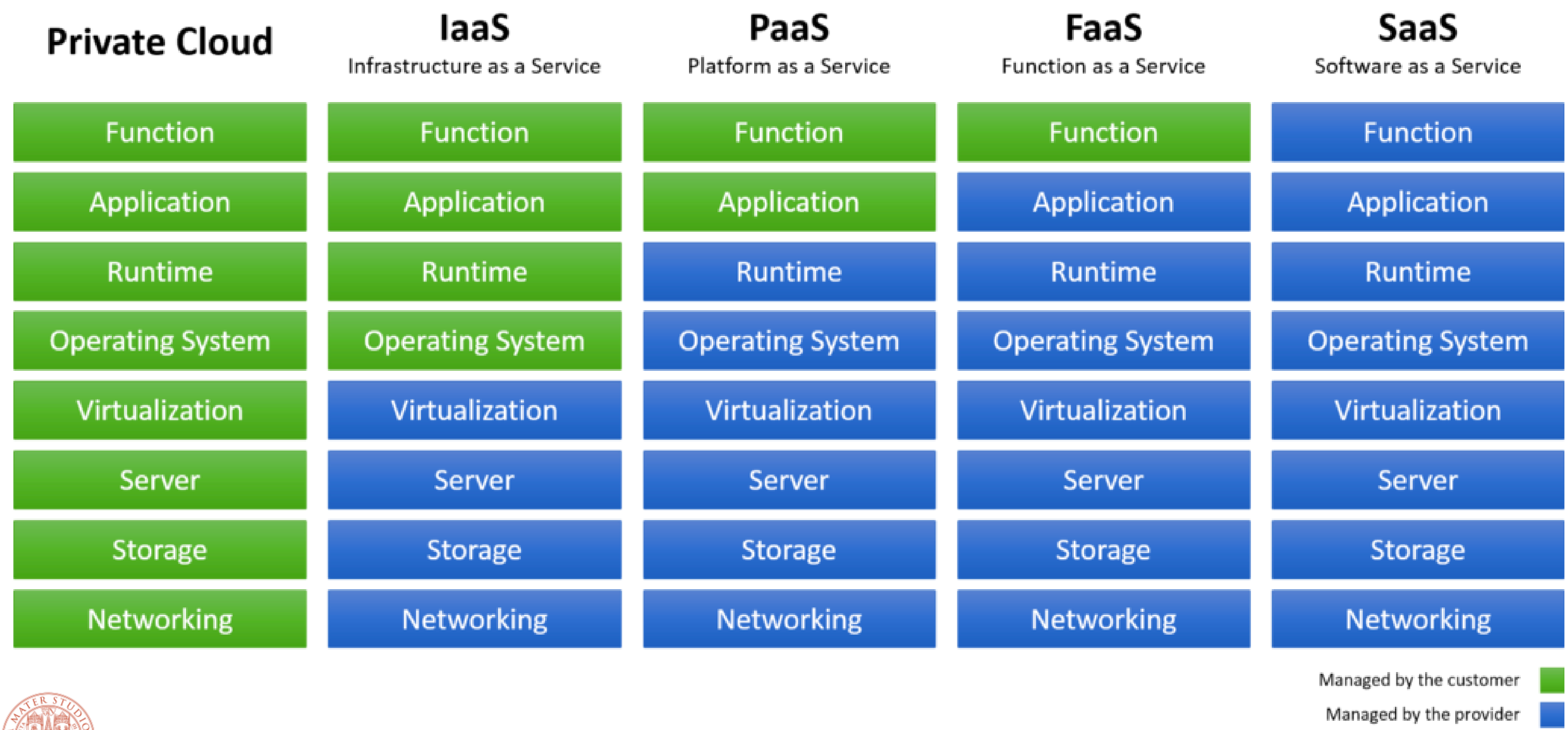
Azure

- Azure is a cloud computing platform developed by Microsoft that offers a really huge number of different services at different levels



Azure services

- **Azure Virtual Machines**, which allow users to create and manage virtual machines in the cloud,
- **Azure Storage**, which provides scalable and durable cloud storage for data.
- **Azure App Service**, which enables users to build and deploy web and mobile apps,
- **Azure Functions**, which provides a serverless computing environment for running code on demand
- **AI and Machine learnings** models
- And many many many …. others

Pricing

- Pay close attention to the costs
- Many services are free **within certain limits**

Windows Virtual Machine
COMPUTE

750 hours
B1S

Create Windows virtual machines with on-demand capacity in seconds. Learn more

Create

Linux Virtual Machine
COMPUTE

750 hours
B1S

Create Linux virtual machines with on-demand capacity in seconds. Learn more

Create

VPN Gateway
NETWORKING/SECURITY

750 hours
VpnGw1 Gateway Type

Establish secure, cross-premises connectivity. Learn more

Create

Azure Blob Storage
STORAGE

5 GB
locally redundant storage (LRS) hot block with 20,000 read and 10,000 write operations

Use massively-scalable object storage for any type of unstructured data. Learn more

Create

Azure Files
STORAGE

100GB
of LRS transaction optimized,k hot, and cool files. 2 million read, list, and other file operations

Migrate to simple, distributed, cross-platform file storage without changing code. Learn more

Create

Key Vault
SECURITY

10,000 transactions
RSA 2048-bit keys or secret operations, Standard tier.

Safeguard and maintain control of keys and other secrets. Learn more

Create

Azure Kubernetes Service (AKS)
COMPUTE

Free

Deploy and manage containers using the tools you choose. Learn more

Create

Azure Database for MySQL
DATABASES

750 hours
of Flexible Server—Burstable B1MS Instance, 32 GB storage, and 32 GB backup storage

Host a fully managed, scalable MySQL database in Azure. Learn more

Create

Azure Database for PostgreSQL
DATABASES

750 hours
of Flexible Server—Burstable B1MS Instance, 32 GB storage, and 32 GB backup storage

Build intelligent, scalable apps with fully managed database for PostgreSQL. Learn more

Create

Azure SQL Database
DATABASES

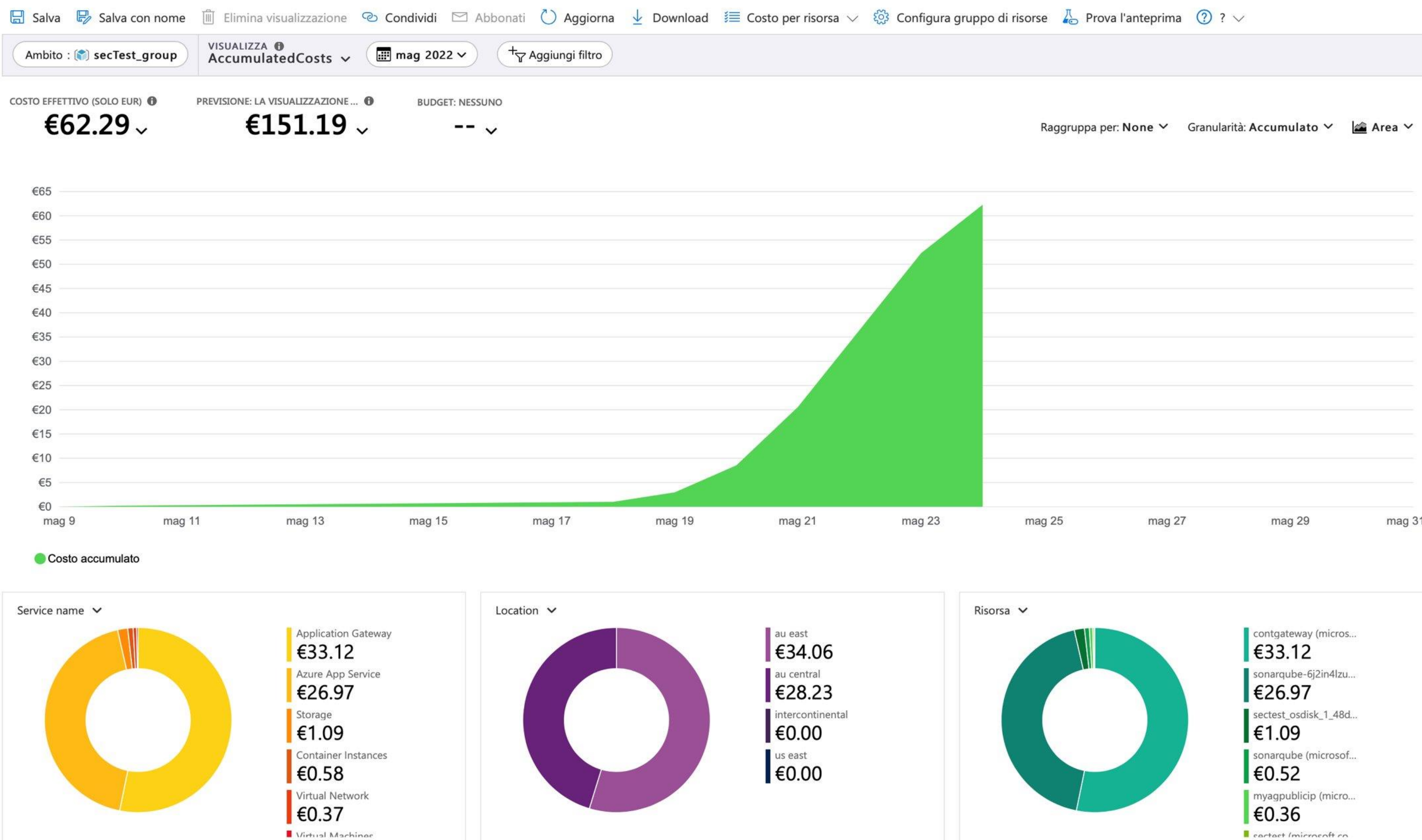
250 GB
S0 instance with 10 database transaction units.

Create a SQL Database that delivers intelligence built-in. Learn more

Create

Pricing

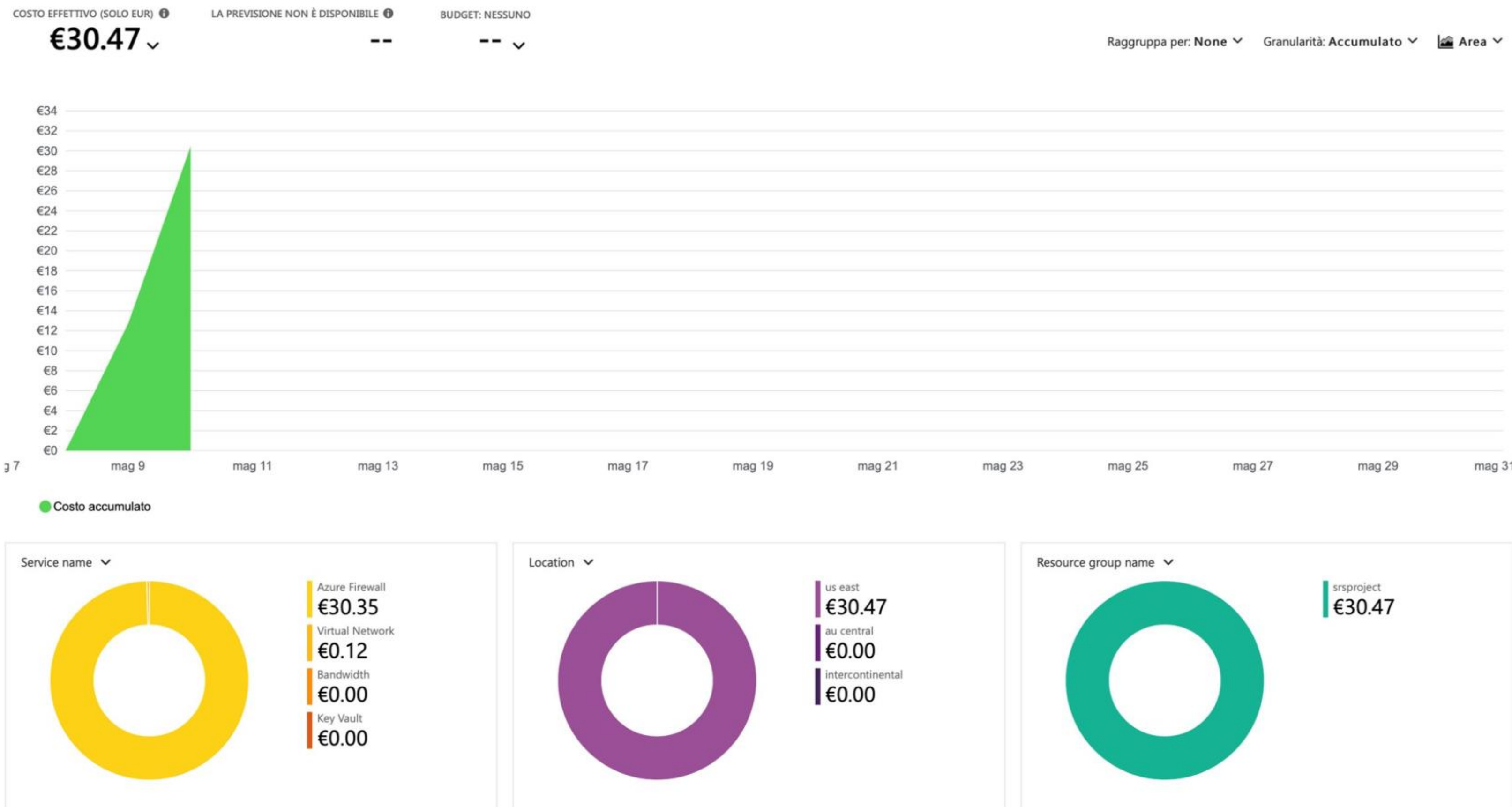
Pay close attention to the costs



Price calculator: https://azure.microsoft.com/en-us/pricing/calculator/

Pricing

- Monitor the cost management service daily
- Plan before



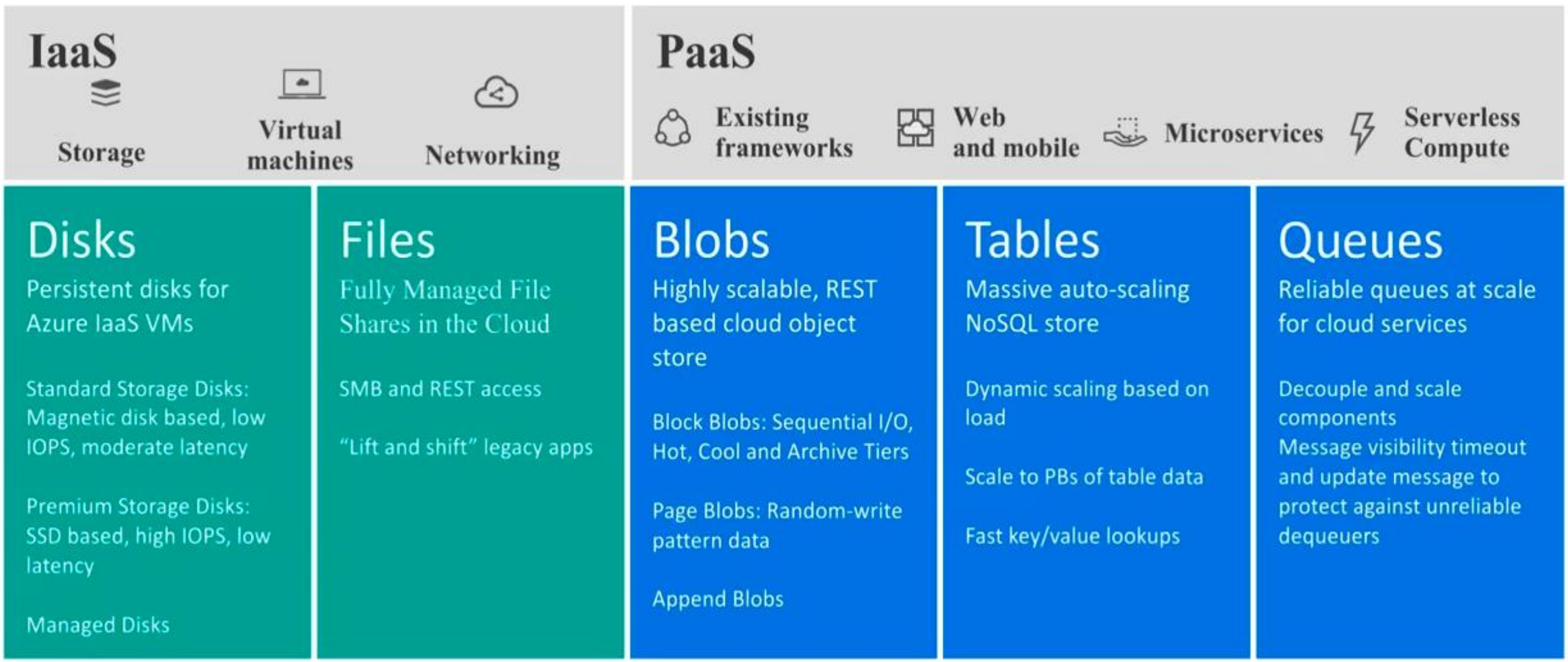
Price calculator: https://azure.microsoft.com/en-us/pricing/calculator/

Virtual Machines

- Azure offer several types of VMs with different computational resources and prices
- You can choose between pre-configured images of operating systems such as **Windows, Linux, and SQL Server** or upload your custom images
- There are also VM **optimized for specific workloads** and performance requirements. You can choose from:
  - general-purpose VMs
  - compute-optimized VMs
  - memory-optimized VMs
  - and GPU-enabled VMs
- They also provides features such as load balancing, automatic scaling, and backup and recovery options

Storage

Azure Storage service offers a range of storage solutions, including Blob storage, Queue storage, File storage, Tables and Disk storage, each designed for specific storage scenarios.



Storage (Disk)

- **Disk storage** is designed for providing persistent block-level storage for virtual machines in the cloud
- Designed for **99.999%** availability
  - **Durability**: Azure Disk Storage is highly durable, with data being replicated across multiple disks in a region to ensure availability
  - **Scalability**: Azure Disk Storage allows you to scale storage as needed, with support for up to 64TB of data per disk
  - **Encryption**: Azure Disk Storage supports encryption at rest
  - **Snapshots**: Azure Disk Storage allows you to take snapshots of disks, providing a point-in-time backup

Storage (File)

File storage: it is designed for storing and sharing files within organizations, providing a fully managed file share in the cloud, used to replace or supplement traditional on-premises file servers or network-attached storage (NAS) devices or as persistent volumes for stateful containers.

- **Scalability**: Azure Files storage can scale up to 5TB per share, making it suitable for large file-based workloads
- **Redundancy**: It is highly redundant, with data being replicated across multiple Azure data centers to ensure data availability
- **Security**:It supports Azure Active Directory (Azure AD) authentication and role-based access control (RBAC), ensuring that only authorized users have access to files

Storage (Blob)

Blob storage is designed for storing unstructured data, such as documents, images, and videos. It provides high scalability, performance, and availability

- It **integrates** with other Azure services such as Azure Functions, Azure Stream Analytics, and Azure Machine Learning to provide a complete solution for data storage and analysis
- It also works with **Azure Content Delivery Network** by caching the most frequently used content, and then storing it on edge servers for fast retrieving
- It organizes data into **containers**, which can be thought of as top-level folders. Within each container, users can create one or more **blobs**, which are the individual data objects being stored
- Optimized storage for big data analytics workloads (*Data Lake*)

Storage (Queue)

Queue storage: is designed for building message-based applications that require reliable and asynchronous communication between components

- **Processing messages asynchronously**: Applications can add messages to a queue for processing later, which can help improve application responsiveness and scalability
- **Decoupling components**: allows for the decoupling of application components, which makes it easier to build highly scalable and fault-tolerant distributed systems
- **Distributed processing:** enables distributed processing of messages across multiple worker roles or instances, allowing for highly parallelized processing of messages

Storage (Cosmos/Table)

- Azure Cosmos DB supports semi-structured or **NoSQL data**
- It provides a **highly scalable and low-latency NoSQL database** that supports multiple data models such as document, key-value, graph, and column-family. Azure Cosmos DB
- Designed to support mission-critical applications that require low latency, high availability, and global distribution
- It is also **ACID** (atomicity, consistency, isolation, durability) compliant, so you can be assured that your transactions are completed according to those strict requirements

Storage (Cosmos/Table)

Azure Cosmos DB for NoSQL (recommended)

Azure Cosmos DB's core, or native API for working with documents. Supports fast, flexible development with familiar SQL query language and client libraries for .NET, JavaScript, Python, and Java.

Create

Azure Cosmos DB for MongoDB

Fully managed database service for apps written for MongoDB. Recommended if you have existing MongoDB workloads that you plan to migrate to Azure Cosmos DB.

Create

Azure Cosmos DB for Apache Cassandra

Fully managed Cassandra database service for apps written for Apache Cassandra. Recommended if you have existing Cassandra workloads that you plan to migrate to Azure Cosmos DB.

Create

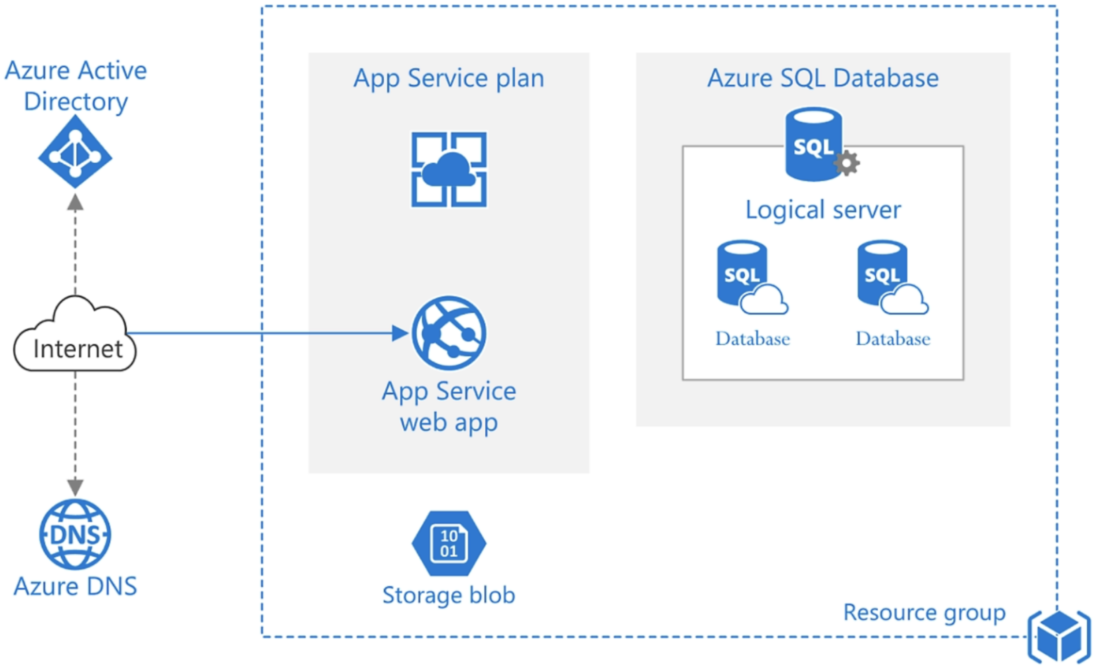
Azure Cosmos DB for PostgreSQL

Fully-managed relational database service for PostgreSQL with distributed query execution, powered by the Citus open source extension. Build new apps on single or multi-node clusters—with support for JSONB, geospatial, rich indexing, and high-performance scale-out.

Create

App service

- Azure App Service is a platform as a service (PaaS) capable of hosting web applications, REST APIs, and mobile back ends
- It provides built-in features such as auto-scaling, load balancing, and high availability, ensuring that applications can handle high traffic and scale as needed



Azure Functions

- **Function-as-a-Service** (**FaaS**) is a kind of cloud computing service that allows developers to build, compute, run, and manage application packages as functions without having to maintain the infrastructure
- Azure Functions is a serverless computing service provided by Microsoft Azure that enables developers to build and run **event-driven** applications and microservices in the cloud
- You can write code in multiple languages such as C#, Java, JavaScript, Python, and PowerShell, and create serverless functions that can be triggered by various events, such as HTTP requests, queue messages, timers, and more

Security Services

- All the previously mentioned services are integrated with the security suite of Azure
- Also, in this case we have a huge number of services that can allows to define fine-grained security policies
- It is crucial to define which are our **priorities** and the most **critical resources** of your service

Microsoft Sentinel
Cloud-native SIEM and intelligent security analytics

Microsoft Defender for Cloud
Extend threat protection to any infrastructure

Application Gateway
Build secure, scalable, highly available web front ends in Azure

Key Vault
Safeguard and maintain control of keys and other secrets

VPN Gateway
Establish secure, cross-premises connectivity

Azure DDoS Protection
Protect your Azure resources from distributed denial-of-service (DDoS) attacks

Azure Bastion
Fully managed service that helps secure remote access to your virtual machines

Web Application Firewall
A cloud-native web application firewall (WAF) service that provides powerful protection for web apps

Azure Firewall
Protect your Azure Virtual Network resources with cloud-native network security

Azure Firewall Manager
Central network security policy and route management for globally distributed, software-defined perimeters

Security services

Azure Cloud Defender

- **CSPM**(Cloud Security Posture Management): help in fixing security issues and monitoring security posture
    - Generate a security score
    - Provides recommendations for advanced security
    - Analyze and secure attack paths
- **CWP**(Cloud Workload Protection Platform): Identify unique workload security requirements
- It is integrated with **Azure Threat Intelligence** service to provide detailed information on the latest threats and vulnerabilities based on the services you are using

Security services

Recommendations ···

Refresh | Download CSV report | Open query | Guides & Feedback

Secure score recommendations | All recommendations

Azure | AWS | GCP

Secure score ⓘ

14%

Active items

Controls 12/15

Recommendations 13/27

Resource health

Unhealthy (3) | Healthy (1) | Not applicable (2)

Search recommendations | Recommendation status == **None** | Severity == **None** | Resource type == **None** | Recommendation maturity == **None** | Add filter

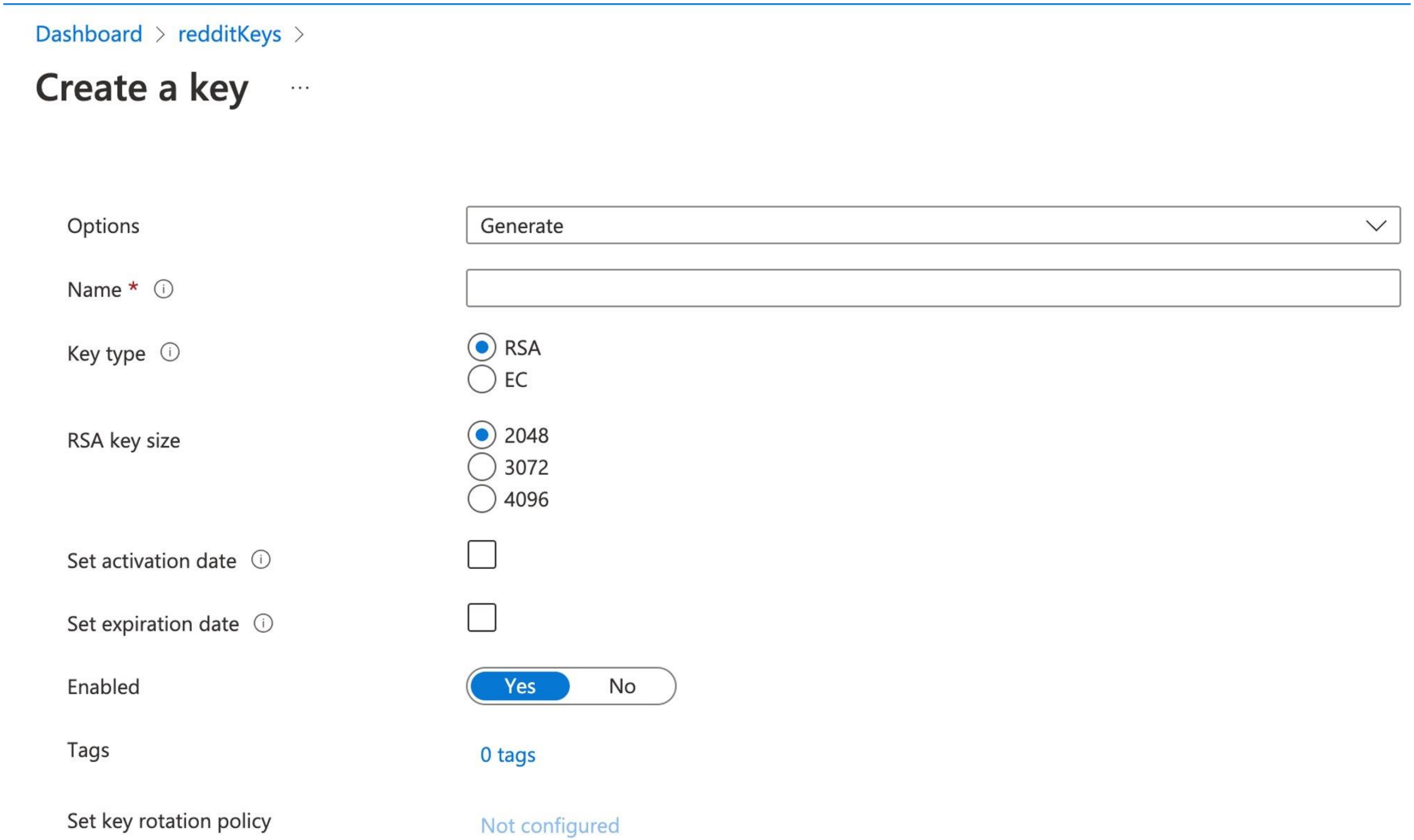
Name	Max score	Current score	Potential score increase	Status	Unhealthy resources	Insights
Enable MFA	10	0.00	+ 18%	Unhealthy	1 of 1 resources	
Secure management ports	8	0.00	+ 14%	Unhealthy	1 of 1 resources	
Remediate vulnerabilities	6	0.00	+ 11%	Unhealthy	1 of 1 resources	
Apply system updates	6	0.00	+ 11%	Unhealthy	1 of 1 resources	
Remediate security configurations	4	0.00	+ 7%	Unhealthy	1 of 1 resources	
Manage access and permissions	4	4.00		Completed	0 of 3 resources	
Restrict unauthorized network access	4	0.00	+ 7%	Unhealthy	1 of 3 resources	
Enable encryption at rest	4	0.00	+ 7%	Unhealthy	1 of 1 resources	
Encrypt data in transit	4	4.00		Completed	0 of 1 resources	
Apply adaptive application control	3	0.00	+ 5%	Unhealthy	1 of 1 resources	
Enable endpoint protection	2	0.00	+ 4%	Unhealthy	1 of 1 resources	
Enable auditing and logging	1	0.00	+ 2%	Unhealthy	1 of 1 resources	
Implement security best practices	Not scored	Not scored		Unhealthy	1 of 6 resources	
Enable enhanced security features	Not scored	Not scored		Unhealthy	1 of 1 resources	
Protect applications against DDoS attacks	Not scored	Not scored		Completed	0 of 1 resources	

Give us feedback

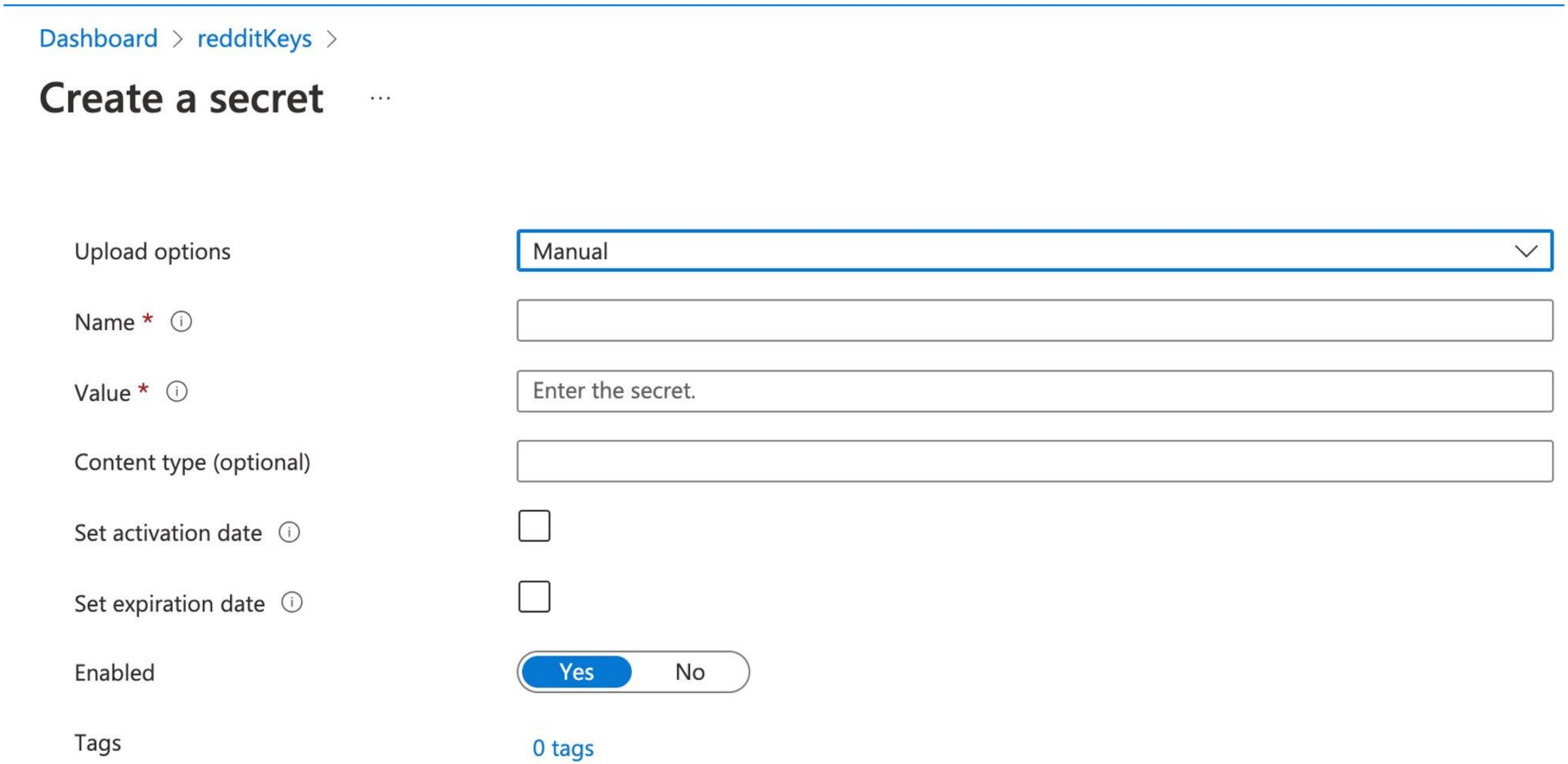
Security services (Secrets management)

- Probably during your project, you need to store some secrets (API tokens, Passwords, certificates, etc.)
- **Azure Key Vault** allows detailed management of secrets and allows you to:
  - **Determine the duration of a secret**, thus allowing you to determine its expiration, forcing its periodic updating
  - **Manage access to the secret with a whitelist approach** by determining the IP addresses from which it can be accessed
  - Create **policies** for a single user or application by defining the level of Authorization and therefore how to be able to interact with the various secrets
  - The entire security system of Azure try to follow a Zero trust approach

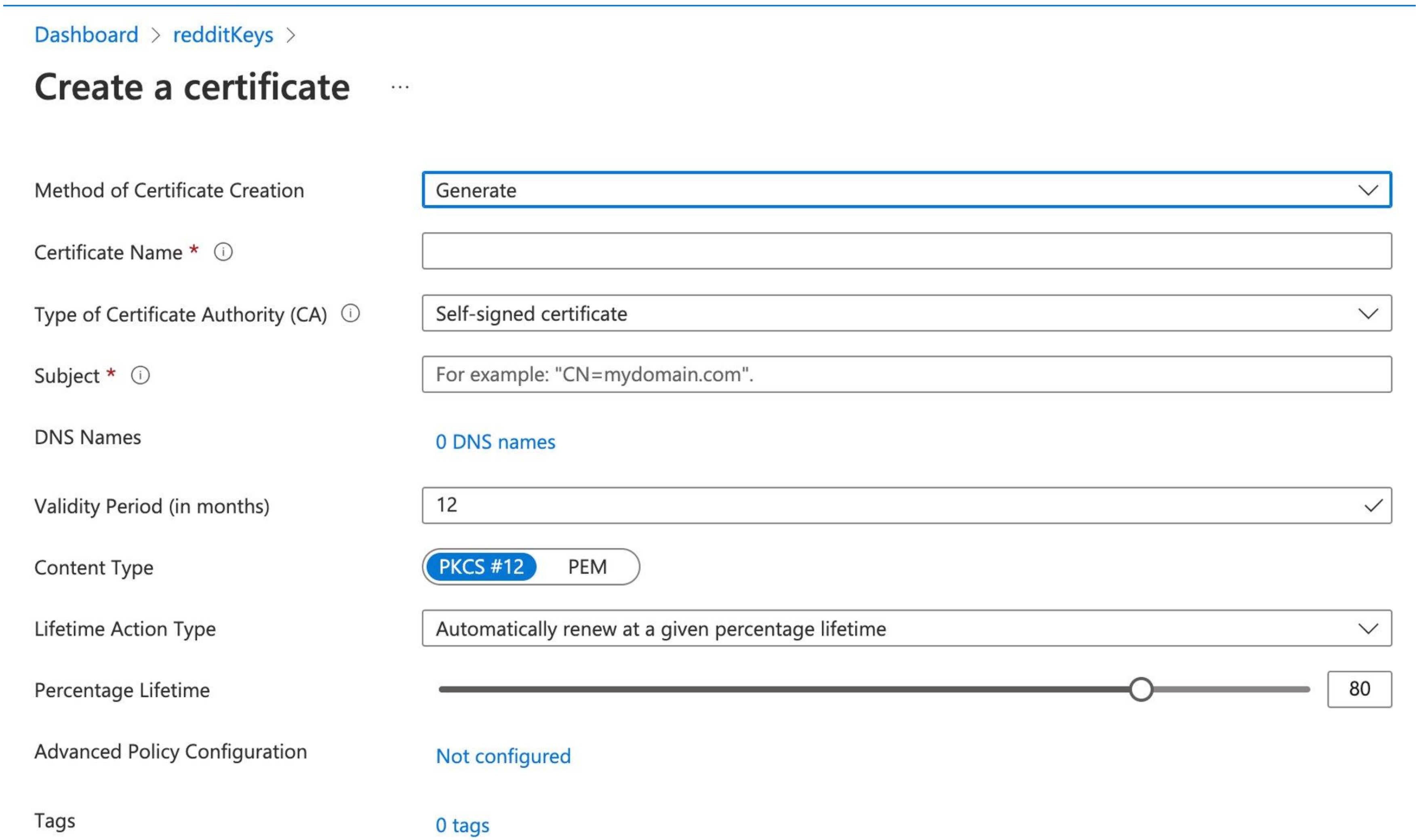
Security services (Secrets management)



Security services (Secrets management)



Security services (Secrets management)

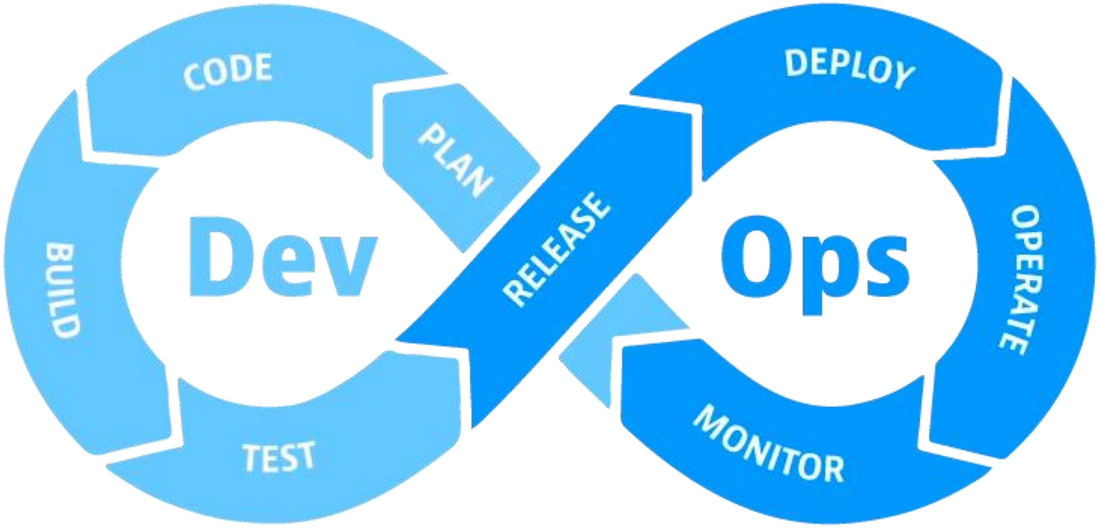


Security services

- Integration with several external security tools for a **continuous Vulnerability Assessment**
- Supporting the **Infrastructure as Code** (IaC) model

DevSecOps tools

- Among the various services azure offers the possibility to use Aure DevOps to implement **CI-CD** tools
- It is a suite of services which simplify the adoption of the *DevSecOps* model



DevSecOps tools

Azure DevOps

Plan smarter, collaborate better, and ship faster with a set of modern dev services.

Start free   Start free with GitHub

Overview   Features   Security   Get started

Azure Boards

Deliver value to your users faster using proven agile tools to plan, track, and discuss work across your teams.

Learn more >

Azure Pipelines

Build, test, and deploy with CI/CD that works with any language, platform, and cloud. Connect to GitHub or any other Git provider and deploy continuously.

Learn more >

Azure Repos

Get unlimited, cloud-hosted private Git repos and collaborate to build better code with pull requests and advanced file management.

Learn more >

Azure Test Plans

Test and ship with confidence using manual and exploratory testing tools.

Learn more >

Azure Artifacts

Create, host, and share packages with your team, and add artifacts to your CI/CD pipelines with a single click.

Learn more >

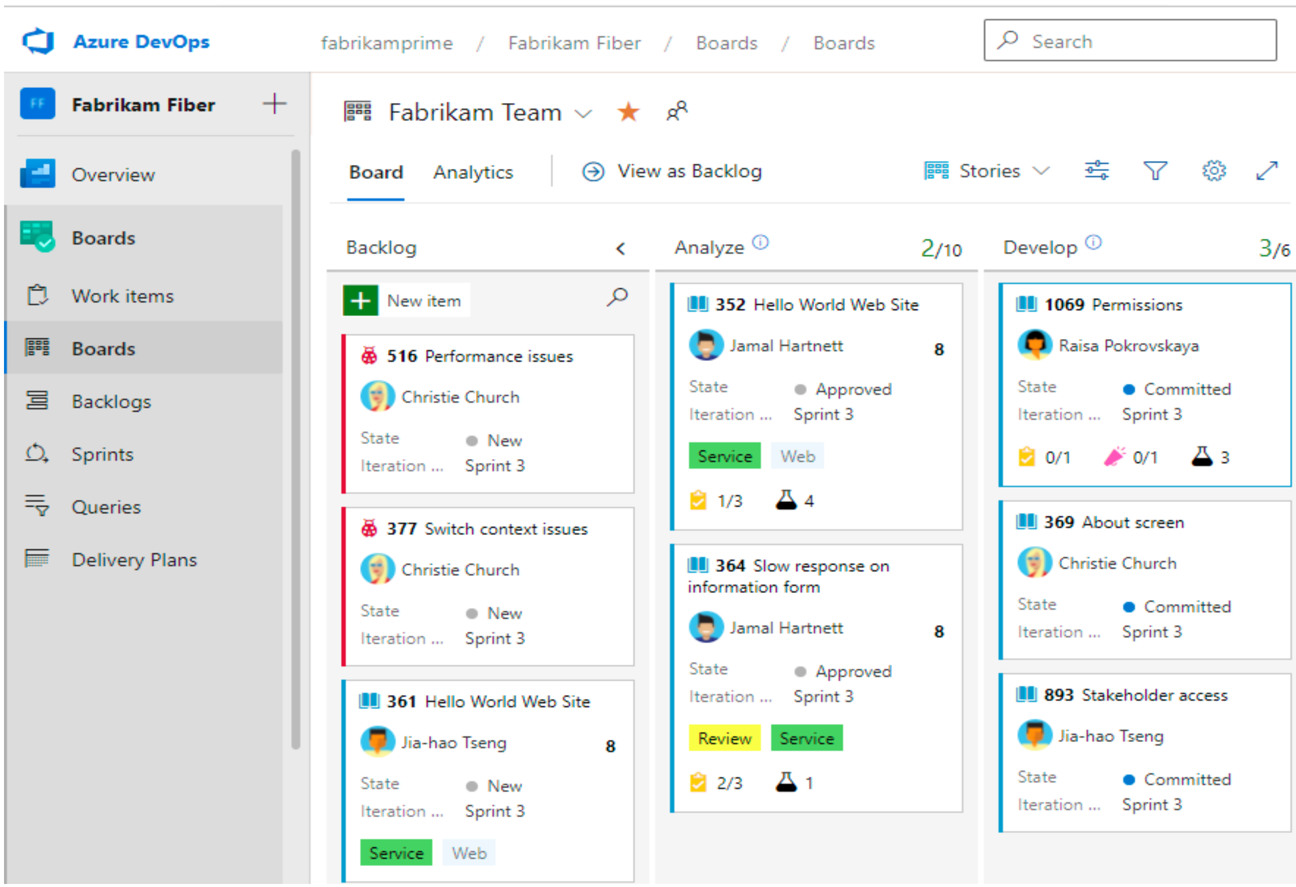
Extensions Marketplace

Access extensions from Slack to SonarCloud to 1,000 other apps and services—built by the community.

Learn more >

DevSecOps tools(Boards)

Azure Boards: A service for **agile** project management that allows you to plan, track, and discuss work across their entire software development process.



DevSecOps tools (Pipeline)

Azure Pipelines (IaC): A continuous integration and continuous deployment (CI/CD) service that automates the building, testing, and deployment of applications to various platforms.

- **Build:** It can automatically build and compile code changes for software projects. It supports builds from source code repositories like GitHub, Bitbucket, and Azure Repos
- **Test:** provides **automated testing capabilities** to ensure that code changes are stable and do not break existing functionality. It supports a variety of testing frameworks and tools for unit tests, integration tests, and acceptance tests.
- **Release:** It automates the deployment of software projects to different environments such as development, staging, and production